Technical Note: Multi-Sphere T-Matrix Theory

Mathematical Formulation and Computational Implementation

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Abstract

This document provides a self-contained description of the mathematical theory behind the Multi-Sphere T-Matrix (MSTM) method as implemented in MSTMforCAS.jl. It covers the VSWF expansion formalism, Lorenz–Mie theory for single spheres, the multi-sphere interaction equation, VSWF translation addition theorem, the Complex BiConjugate Gradient (CBICG) iterative solver, the FFT-accelerated translation algorithm, and the output quantities (scattering amplitudes and cross-section efficiencies). The implementation follows the formulation of Mackowski (1991, 1994, 1996), Xu (1996), Mackowski & Mishchenko (2011), and Mackowski & Kolokolova (2022). Scattering amplitude conventions follow Bohren & Huffman (1983, hereafter BH83).

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1 Problem Setup

1.1 Geometry and incident field

Consider N_S homogeneous, non-overlapping spheres embedded in an infinite, lossless, homogeneous medium of real refractive index n_{med}. Sphere i ($i = 1, \ldots, N_S$) has centre position $\boldsymbol{r}_i$, radius a_i, and complex refractive index m_{sphere} (common to all spheres in the parameter-sweep mode of this code). A time-harmonic plane wave ($e^{-i\omega t}$ convention, following BH83) propagates along the $+z$ axis in the medium with vacuum wavelength λ_0 and wavenumber in the medium

$$k = \frac{2\pi\, n_{\text{med}}}{\lambda_0}. \tag{1}$$

A positive imaginary part of m_{sphere} corresponds to absorption. All lengths are non-dimensionalised by k: positions become $k\boldsymbol{r}_i$, radii become size parameters $x_i = ka_i$, and the relative refractive index entering all scattering formulas is $m_{\text{rel}} = m_{\text{sphere}}/n_{\text{med}}$. Tildes are suppressed hereafter; all equations are in dimensionless units unless stated otherwise.

The volume-equivalent sphere radius a_{eff} and its size parameter x_{eff} are defined as

$$a_{\text{eff}} = \left(\sum_{i=1}^{N_S} a_i^3 \right)^{1/3}, \qquad x_{\text{eff}} = k\, a_{\text{eff}}. \tag{2}$$

These appear in the normalisation of cross-section efficiencies (Sec. 5).

1.2 Vector spherical wave function expansion

The electromagnetic field in each region is expanded in vector spherical wave functions (VSWFs) $\boldsymbol{M}_{mn}^{(\nu)}$, $\boldsymbol{N}_{mn}^{(\nu)}$ of degree m and order n ($n = 1, 2, \ldots$; $m = -n, \ldots, n$), where $\nu = 1$ denotes regular (Bessel j_n) waves and $\nu = 3$ denotes outgoing (Hankel $h_n^{(1)}$) waves. The normalisation convention for these VSWFs follows that of Mishchenko et al. (2002) and Mackowski (1994). The scattered field from sphere i is expanded in outgoing VSWFs centred at $\boldsymbol{r}_i$:

$$\boldsymbol{E}_{\text{sca}}^{(i)}(\boldsymbol{r}) = \sum_{n=1}^{N_i} \sum_{m=-n}^{n} \left[a_{mn}^{(i)}\, \boldsymbol{M}_{mn}^{(3)}(\boldsymbol{r} - \boldsymbol{r}_i) + b_{mn}^{(i)}\, \boldsymbol{N}_{mn}^{(3)}(\boldsymbol{r} - \boldsymbol{r}_i) \right] \tag{3}$$

where N_i is the truncation order for sphere i. The field exciting sphere i — the incident plane wave plus the scattered fields from all other spheres — is likewise expanded in regular VSWFs $\boldsymbol{M}_{mn}^{(1)}, \boldsymbol{N}_{mn}^{(1)}$ centred at $\boldsymbol{r}_i$.

1.3 Left–right circular (lr_tran) basis

Following Mackowski (1996, §2), the code works in the left–right circular VSWF basis obtained by the linear combinations

$$\tilde{\boldsymbol{N}}_{mn,L}^{(\nu)} = \boldsymbol{N}_{mn,1}^{(\nu)} + \boldsymbol{N}_{mn,2}^{(\nu)}, \qquad \tilde{\boldsymbol{N}}_{mn,R}^{(\nu)} = \boldsymbol{N}_{mn,1}^{(\nu)} - \boldsymbol{N}_{mn,2}^{(\nu)} \tag{4}$$

where the subscripts 1, 2 refer to the TE and TM modes. The fundamental mathematical advantage of this basis is that it completely decouples the translation matrix $\mathbf{H}_{ij}$ between the two polarisation blocks. While the single-sphere T-matrix is diagonal in the combined multipole index $mn = n(n+1) + m$ in both the standard TE/TM and lr_tran bases, in the lr_tran basis it couples the two polarisation blocks (labeled $p = 1$ for left and $p = 2$ for right). This structure greatly simplifies the multi-sphere interaction equations, as the computationally expensive translation operations can be performed independently for each polarisation block.

2 Single-Sphere Scattering: Lorenz–Mie Theory

For a homogeneous sphere with size parameter x and relative refractive index m_{rel}, the Lorenz–Mie scattering coefficients a_n (TM/electric) and b_n (TE/magnetic) are given by (BH83, Eq. 4.88):

$$a_n = \frac{\left[\dfrac{D_n(m_{\mathrm{rel}}\,x)}{m_{\mathrm{rel}}} + \dfrac{n}{x}\right]\psi_n(x) - \psi_{n-1}(x)}{\left[\dfrac{D_n(m_{\mathrm{rel}}\,x)}{m_{\mathrm{rel}}} + \dfrac{n}{x}\right]\xi_n(x) - \xi_{n-1}(x)} \tag{5}$$

$$b_n = \frac{\left[m_{\mathrm{rel}}\,D_n(m_{\mathrm{rel}}\,x) + \dfrac{n}{x}\right]\psi_n(x) - \psi_{n-1}(x)}{\left[m_{\mathrm{rel}}\,D_n(m_{\mathrm{rel}}\,x) + \dfrac{n}{x}\right]\xi_n(x) - \xi_{n-1}(x)} \tag{6}$$

where:

- $\psi_n(z) = z\,j_n(z)$ is the Riccati–Bessel function (j_n: spherical Bessel function of the first kind)
- $\xi_n(z) = z\,h_n^{(1)}(z) = \psi_n(z) + i\,\chi_n(z)$ is the Riccati–Hankel function
- $\chi_n(z) = -z\,y_n(z)$ (y_n: spherical Bessel function of the second kind)
- $D_n(z) = \frac{d}{dz}\ln[z\,j_n(z)] = \psi_n'(z)/\psi_n(z)$ is the logarithmic derivative

The logarithmic derivative $D_n(y)$ with $y = m_{\mathrm{rel}}\,x$ is computed by downward recurrence (Wiscombe 1980):

$$D_{n-1}(y) = \frac{n}{y} - \frac{1}{D_n(y) + n/y}, \tag{7}$$

starting from a sufficiently large $n_{\mathrm{start}} \geq \max(N_{\mathrm{max}}, |y|) + 16$ where $D_{n_{\mathrm{start}}} \approx 0$. This recurrence is unconditionally stable for any complex y. The Riccati–Bessel functions $\psi_n(x)$ and $\chi_n(x)$ (for real argument x) are computed by upward recurrence:

$$f_n(x) = \frac{2n-1}{x}\,f_{n-1}(x) - f_{n-2}(x) \tag{8}$$

with starting values $\psi_0 = \sin x$, $\psi_1 = \sin x/x - \cos x$, $\chi_0 = -\cos x$, $\chi_1 = -\cos x/x - \sin x$.

The multipole expansion is truncated at order N_i determined by the criterion of Wiscombe (1980) as implemented in the MSTM code:

$$N_i = \mathrm{round}\Big(x_i + 4\,x_i^{1/3}\Big) + 5, \tag{9}$$

refined by a convergence check: starting from this upper bound, N_i is reduced until the cumulative partial extinction efficiency $Q_{\mathrm{ext}}^{(N_i)}$ deviates from the total by more than 10^{-6}.

In the lr_tran basis, the single-sphere T-matrix for multipole order n takes the 2×2 form coupling the $p = 1$ and $p = 2$ blocks (Mackowski 1996, derived from the `mieoa` routine in MSTM v4.0):

$$\mathbf{T}_n = -\frac{1}{2}\begin{pmatrix} a_n + b_n & a_n - b_n \\ a_n - b_n & a_n + b_n \end{pmatrix} \tag{10}$$

This matrix is diagonal in the combined index mn (it does not mix different (m, n) pairs) but couples the two circular polarisation blocks. For an isotropic sphere, $a_n \neq b_n$ in general, so the off-diagonal coupling is physically meaningful and arises from the difference in electric and magnetic multipole scattering.

3 Multi-Sphere Interaction Equation

3.1 Coupled field equations

The field that excites sphere i consists of the external incident wave plus the scattered fields from all other spheres, translated to the coordinate origin of sphere i using the VSWF translation addition

theorem (Mackowski 1991, 1996; Xu 1996):

$$\boldsymbol{p}_{\text{exc}}^{(i)} = \boldsymbol{p}_{\text{inc}}^{(i)} + \sum_{j \neq i} \mathbf{H}_{ij}\, \boldsymbol{a}^{(j)} \tag{11}$$

where $\mathbf{H}_{ij}$ is the translation matrix that re-expands outgoing VSWFs centred at $\boldsymbol{r}_j$ into regular VSWFs centred at $\boldsymbol{r}_i$ (Hankel-to-Bessel type), and $\boldsymbol{a}^{(j)}$ is the vector of scattered-field expansion coefficients of sphere j.

The T-matrix relates the exciting field to the scattered field:

$$\boldsymbol{a}^{(i)} = \mathbf{T}_i\, \boldsymbol{p}_{\text{exc}}^{(i)}. \tag{12}$$

Substituting and rearranging into a single global linear system for $\boldsymbol{x} = (\boldsymbol{a}^{(1)}, \ldots, \boldsymbol{a}^{(N_S)})^\top$ gives (Mackowski 1994, Eq. 6):

$$(\mathbf{I} - \mathbf{T}\,\mathbf{A})\, \boldsymbol{x} = \mathbf{T}\, \boldsymbol{p}_{\text{inc}} \tag{13}$$

where $\mathbf{T} = \text{diag}(\mathbf{T}_1, \ldots, \mathbf{T}_{N_S})$ is the block-diagonal T-matrix and $\mathbf{A}$ is the off-diagonal translation operator with blocks $\mathbf{H}_{ij}$ $(i \neq j)$ and zeros on the diagonal. The dimension of this system is $\sum_i 2\, N_i(N_i + 2)$ — twice the multipole block size because each sphere carries both $p = 1$ and $p = 2$ polarisation blocks.

3.2 Translation matrix via rotation–axial translation–rotation

The translation matrix $\mathbf{H}_{ij}$ is computed via the factorisation into rotation, axial translation, and inverse rotation (Mackowski 1996, §3; Xu 1996, §4), which avoids the direct evaluation of Gaunt integrals and is more efficient:

1. **Rotation.** Compute the spherical coordinates $(d_{ij}, \theta_{ij}, \phi_{ij})$ of the translation vector $\boldsymbol{r}_i - \boldsymbol{r}_j$. The azimuthal phase factors $e^{im\phi_{ij}}$ and the normalised associated Legendre functions $P_n^m(\cos\theta_{ij})$ provide the rotation from the laboratory frame to a frame where the translation vector is along z.
2. **Axial translation coefficients.** In the rotated frame where the two origins are separated along z by distance d_{ij}, the translation coefficients factorise. The key quantity involves a sum over intermediate order w:

$$C_{mn,kl} = \sum_{w=w_{\min}}^{n+l} c_w(m,n,k,l)\; h_w^{(1)}(d_{ij})\; P_w^{m-k}(\cos\theta_{ij}) \tag{14}$$

   where $w_{\min} = \max(|n-l|, |m-k|)$ and $c_w(m,n,k,l)$ are products of vector coupling (Gaunt) coefficients. These coupling coefficients are related to Clebsch–Gordan coefficients and are evaluated using a combined downward–upward recurrence algorithm (Xu 1996, §3; Mackowski 1996, Appendix) that avoids explicit factorial computation and maintains numerical stability even at high multipole orders.
3. **Polarisation coupling in the lr_tran basis.** In this basis the translation matrix has a particularly simple structure: the even-w and odd-w contributions (relative to $n + l$) combine to give the two polarisation blocks independently. Specifically, denoting the even- and odd-w partial sums as A and B:

$$H_{kl,mn}^{(p=1)} = e^{i(m-k)\phi_{ij}}\,(A + iB)\,, \qquad H_{kl,mn}^{(p=2)} = e^{i(m-k)\phi_{ij}}\,(A - iB)\,. \tag{15}$$

   This decoupling is the fundamental advantage of the lr_tran basis (Mackowski 1996, Eq. 18): the two circular polarisation modes interact with each other only through the T-matrix, not through translation.

The coupling coefficients are precomputed once and cached as a 3D array `tran_coef[mn, kl, w]` (Fortran routine `gentrancoefconstants`), while the distance-dependent Hankel functions and Legendre functions are evaluated per sphere pair.

3.3 FFT-accelerated translation

For large numbers of spheres ($N_S \gtrsim 100$), the direct pairwise evaluation of $\mathbf{A}\,\boldsymbol{v}$ scales as $O(N_S^2)$ per iteration and dominates the computation time. Mackowski & Kolokolova (2022) introduced a discrete Fourier convolution (DFC) scheme into MSTM v4.0 that reduces this to approximately $O(N_S \ln N_S)$ scaling. The algorithm proceeds as follows:

1. **Grid construction.** A uniform cubic grid with cell spacing d is mapped onto the target volume. The cell spacing is chosen from the mean monomer radius and a target volume fraction f_v as $d = \bar{a}\,(4\pi/3f_v)^{1/3}$, and each grid dimension is rounded up to the nearest integer of the form $2^a \times 3^b \times 5^c$ for efficient FFT computation (Temperton's generalised prime-factor FFT).
2. **Sphere–node association.** Each sphere is assigned to the grid cell containing its centre. A *neighbour hole* region is defined around each cell: for all cell pairs within this exclusion region (typically $|i-j| \le \sqrt{2}$, i.e., self + face-adjacent + edge-adjacent cells), sphere–sphere interactions are computed by direct Hankel-type translation, preserving accuracy at short inter-sphere distances.
3. **Sphere → Node.** For each sphere, the scattered-field expansion coefficients are translated to its associated grid node using a regular (Bessel j_n) translation matrix $\mathbf{J}$. This is valid because the sphere centre is within the grid cell (distance less than $d/2$), well within the convergence region of the regular addition theorem.
4. **Node-to-node convolution.** The grid-based translation operator is a discrete convolution: the translation matrix from node $\boldsymbol{i}$ to node $\boldsymbol{j}$ depends only on the displacement $\boldsymbol{j}-\boldsymbol{i}$. This convolution is evaluated in the frequency domain:

$$\hat{g}_\mu(\boldsymbol{k}) = \sum_\nu \hat{H}_{\mu\nu}(\boldsymbol{k})\,\hat{a}_\nu(\boldsymbol{k}) \tag{16}$$

   where μ and ν represent the combined multipole indices. $\hat{H}_{\mu\nu}(\boldsymbol{k})$ is the pre-computed 3D FFT of the cell-to-cell translation matrix (excluding the neighbour-hole cells), $\hat{a}_\nu(\boldsymbol{k})$ is the FFT of the zero-padded node coefficient field, and the product is taken element-wise per index pair (μ,ν). The use of a doubled grid ($2N_x \times 2N_y \times 2N_z$) with zero-padding avoids circular convolution artefacts.
5. **Node → Sphere.** The resulting node-level exciting field is translated back to each sphere position via the reverse regular translation.
6. **Near-field correction.** Direct sphere-to-sphere Hankel-type translation is applied for all sphere pairs within the neighbour-hole region (step 2), adding these contributions to the FFT-based far-field result.

The total cost per iteration becomes $O(N_S + M\log M)$ where M is the number of active grid cells, compared with $O(N_S^2 N_{\max}^4)$ for the direct algorithm. The memory overhead consists of the pre-FFT'd translation matrices, which scale as $O(M \times N_{\text{node}}^2)$ where N_{node} is the node multipole order. Mackowski & Kolokolova (2022) report speed improvements exceeding two orders of magnitude for systems of 10^3–10^4 spheres.

4 Iterative Solution: Complex BiConjugate Gradient

The linear system $\mathbf{L}\,\boldsymbol{x} = \boldsymbol{b}$ where $\mathbf{L} = \mathbf{I} - \mathbf{T}\,\mathbf{A}$ and $\boldsymbol{b} = \mathbf{T}\,\boldsymbol{p}_{\text{inc}}$ is solved by the Complex BiConjugate Gradient (CBICG) method (Mackowski & Mishchenko 2011, §2.3).

4.1 Operator definitions

The CBICG iteration requires both the forward operator $\mathbf{L}\,\boldsymbol{v} = \boldsymbol{v} - \mathbf{T}\,\mathbf{A}\,\boldsymbol{v}$ and the adjoint (conjugate transpose) operator $\mathbf{L}^\dagger\,\boldsymbol{v} = \boldsymbol{v} - \mathbf{A}^\dagger\,\mathbf{T}^\dagger\,\boldsymbol{v}$. The forward operator is computed by applying the translation $\mathbf{A}$ (direct or FFT-accelerated), followed by the block-diagonal T-matrix application $\mathbf{T}$.

The adjoint of the translation operator $\mathbf{A}$ is the computationally expensive part. Mackowski & Mishchenko (2011) exploit the S-parity symmetry of the VSWF translation matrix to avoid explicitly

forming $\mathbf{A}^\dagger$:

$$\mathbf{A}^\dagger = (\mathbf{S}\,\mathbf{A}\,\mathbf{S})^* \tag{17}$$

where $\mathbf{S}$ is the parity operator acting on each sphere's coefficient block as $a_{m,n,p} \mapsto (-1)^m\, a_{-m,n,p}$, and * denotes element-wise complex conjugation. This identity arises from the spatial inversion symmetry and reciprocity of the electromagnetic translation operator. The adjoint translation is evaluated by: (i) applying $\mathbf{S}$, (ii) performing the same forward translation $\mathbf{A}$, (iii) applying $\mathbf{S}$ again, and (iv) complex-conjugating. This reuses the forward translation code path entirely, including the FFT-accelerated version, at no additional implementation cost.

The system is solved independently for two orthogonal incident linear polarisations ($q = 1$ corresponding to x-polarisation, and $q = 2$ corresponding to y-polarisation). The default convergence tolerance is $\varepsilon = 10^{-6}$.

4.2 CBICG algorithm (default solver)

The CBICG (Complex BiConjugate Gradient) algorithm is summarised in Table 1. Each iteration requires one forward operator application $\mathbf{L}\,\boldsymbol{v}$ and one adjoint application $\mathbf{L}^\dagger\,\boldsymbol{v}$, totalling 2 translation operator ($\mathbf{A}$) evaluations per iteration.

Table 1: CBICG algorithm (Mackowski & Mishchenko 2011).

Input: $\mathbf{L}$, $\boldsymbol{b} = \mathbf{T}\,\boldsymbol{p}_{\text{inc}}$, tolerance ε, max iterations $n_{\max}$.

Initialise: $\boldsymbol{x}_0 = \boldsymbol{b}$, $\boldsymbol{r}_0 = \boldsymbol{b} - \mathbf{L}\,\boldsymbol{x}_0$, $\boldsymbol{q}_0 = \overline{\boldsymbol{r}_0}$, $\boldsymbol{p}_0 = \boldsymbol{r}_0$, $\boldsymbol{w}_0 = \boldsymbol{q}_0$, $s_0 = \boldsymbol{q}_0^{\mathsf{H}}\boldsymbol{r}_0$.

For $n = 0, 1, \ldots, n_{\max} - 1$:

1. $\boldsymbol{u} = \mathbf{L}\,\boldsymbol{p}_n$ (1 forward operator application)
2. $\boldsymbol{v} = \mathbf{L}^\dagger\,\boldsymbol{w}_n$ (1 adjoint operator application via S-parity)
3. $\alpha_n = s_n \,/\, \boldsymbol{w}_n^{\mathsf{H}}\boldsymbol{u}$
4. $\boldsymbol{x}_{n+1} = \boldsymbol{x}_n + \alpha_n\,\boldsymbol{p}_n$
5. $\boldsymbol{r}_{n+1} = \boldsymbol{r}_n - \alpha_n\,\boldsymbol{u}$, $\quad \boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \overline{\alpha}_n\,\boldsymbol{v}$
6. $s_{n+1} = \boldsymbol{q}_{n+1}^{\mathsf{H}}\boldsymbol{r}_{n+1}$
7. If $\mathrm{Re}(\boldsymbol{r}_{n+1}^\top\boldsymbol{r}_{n+1})/\|\boldsymbol{b}\|^2 < \varepsilon$: **stop**, return $\boldsymbol{x}_{n+1}$.
8. $\beta_n = s_{n+1}/s_n$
9. $\boldsymbol{p}_{n+1} = \boldsymbol{r}_{n+1} + \beta_n\,\boldsymbol{p}_n$, $\quad \boldsymbol{w}_{n+1} = \boldsymbol{q}_{n+1} + \overline{\beta}_n\,\boldsymbol{w}_n$

Note: the convergence criterion (step 7) uses the *bilinear* (not sesquilinear) inner product $\boldsymbol{r}^\top\boldsymbol{r} = \sum_i r_i^2$, following the Fortran MSTM implementation. The shadow residual $\boldsymbol{q} = \overline{\boldsymbol{r}}$ ensures that $\boldsymbol{q}^{\mathsf{H}}\boldsymbol{r} = \sum_i r_i^2$.

4.3 GMRES algorithm (alternative solver)

As an alternative to CBICG, a GMRES (Generalized Minimal Residual) solver (Saad & Schultz 1986) is available via Krylov.jl (`solver=:gmres`). GMRES requires only the forward operator $\mathbf{L}\,\boldsymbol{v}$ (no adjoint) and minimises the 2-norm of the residual over a Krylov subspace

$$\mathcal{K}_k(\mathbf{L}, \boldsymbol{r}_0) = \mathrm{span}\{\boldsymbol{r}_0,\, \mathbf{L}\,\boldsymbol{r}_0,\, \mathbf{L}^2\,\boldsymbol{r}_0,\, \ldots,\, \mathbf{L}^{k-1}\,\boldsymbol{r}_0\}. \tag{18}$$

The algorithm is summarised in Table 2.

Key properties of GMRES compared to CBICG:

- 1 operator application per iteration (no adjoint), versus 2 for CBICG.
- Monotonically decreasing residual: $\|\boldsymbol{r}_k\|_2 \le \|\boldsymbol{r}_{k-1}\|_2$ by construction.

Table 2: GMRES algorithm (Saad & Schultz 1986).

Input: $\mathbf{L}$, $\boldsymbol{b}$, tolerance ε, restart parameter m, max iterations $n_{\max}$.

Initialise: $\boldsymbol{x}_0 = \boldsymbol{b}$, $\quad \boldsymbol{r}_0 = \boldsymbol{b} - \mathbf{L}\,\boldsymbol{x}_0$, $\quad \beta = \|\boldsymbol{r}_0\|_2$, $\quad \boldsymbol{v}_1 = \boldsymbol{r}_0/\beta$.

For $j = 1, 2, \ldots, \min(m, n_{\max})$:

1. $\boldsymbol{w} = \mathbf{L}\,\boldsymbol{v}_j$ (1 forward operator application)
2. **For** $i = 1, \ldots, j$: $\quad h_{ij} = \boldsymbol{v}_i^{\mathsf{H}}\,\boldsymbol{w}$, $\quad \boldsymbol{w} \leftarrow \boldsymbol{w} - h_{ij}\,\boldsymbol{v}_i$ (modified Gram–Schmidt)
3. $h_{j+1,j} = \|\boldsymbol{w}\|_2$, $\quad \boldsymbol{v}_{j+1} = \boldsymbol{w}/h_{j+1,j}$
4. Solve $\boldsymbol{y}_j = \arg\min_{\boldsymbol{y}} \|\beta\,\boldsymbol{e}_1 - \bar{\mathbf{H}}_j\,\boldsymbol{y}\|_2$ via Givens rotations
5. If $\|\beta\,\boldsymbol{e}_1 - \bar{\mathbf{H}}_j\,\boldsymbol{y}_j\|_2/\|\boldsymbol{b}\|_2 < \varepsilon$: **stop**.

Solution update: $\boldsymbol{x} = \boldsymbol{x}_0 + \mathbf{V}_j\,\boldsymbol{y}_j$, where $\mathbf{V}_j = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j]$.

If $j = m$ and not converged: restart with $\boldsymbol{x}_0 \leftarrow \boldsymbol{x}$.

- Memory: $O(kn)$ for k Krylov basis vectors of length n ($\sim$14 MB for $k = 30$, $n = 48000$).
- The implementation uses $m = 30$ (Krylov.jl `memory` parameter).

Performance comparison. For the MSTM problem class, GMRES requires approximately twice as many iterations as CBICG to reach the same residual tolerance, because the CBICG adjoint operator (via S-parity, (17)) provides additional directional information. The total number of $\mathbf{A}$-operator applications is therefore similar (CBICG: k iterations $\times$ 2; GMRES: $\sim 2k$ iterations $\times$ 1), and the wall-clock performance is comparable. The default solver remains CBICG, consistent with the Fortran MSTM reference implementation.

4.4 Continuation method for refractive index sweeps

When sweeping the complex refractive index m over a grid while keeping the aggregate geometry fixed, the solution $\boldsymbol{x}(m)$ of (13) varies continuously with m. A natural acceleration strategy is to use the converged solution $\boldsymbol{x}(m_j)$ at grid point m_j as the initial guess $\boldsymbol{x}_0$ for the iterative solver at the neighbouring grid point m_{j+1}, instead of the default initial guess $\boldsymbol{x}_0 = \boldsymbol{b} = \mathbf{T}\,\boldsymbol{p}_{\text{inc}}$. This is known as a *continuation method* (or warm-start).

When the refractive index changes from m_j to m_{j+1}, only the Mie coefficients a_n, b_n (and hence the T-matrix $\mathbf{T}$) change; the translation operator $\mathbf{A}$ depends only on geometry and is unchanged. For small $|m_{j+1} - m_j|$, the new solution is close to the previous one, so the initial residual $\|\boldsymbol{b}_{j+1} - \mathbf{L}_{j+1}\,\boldsymbol{x}(m_j)\|$ is much smaller than $\|\boldsymbol{b}_{j+1} - \mathbf{L}_{j+1}\,\boldsymbol{b}_{j+1}\|$, leading to fewer iterations.

The continuation is applied automatically in the parameter sweep (`run_parameter_sweep`): within each (aggregate, medium condition) group, the solver coefficients $\boldsymbol{x}$ from the previous refractive index grid point are passed as the initial guess to the next. If the equation system size changes (e.g., due to a change in the converged Mie truncation order), the continuation is skipped and the default initial guess is used.

Benchmarks (1000 spheres, FFT mode, $\Delta m \approx 0.05$) show a reduction from 15 to 9–11 iterations per grid point, corresponding to a wall-time speedup of approximately 1.2$\times$. For smaller aggregates ($N_p \sim 100$) where the cold-start solver already converges in 3–4 iterations, the improvement is marginal.

Snake-path grid traversal. The effectiveness of the continuation method depends critically on the step size $|m_{j+1} - m_j|$ between consecutive grid points. For a 2-D grid of $N_r \times N_i$ points spanning the real and imaginary parts of m, a naive row-major (raster) traversal sweeps m_{imag} from min to max for each value of m_{real}, then resets to the minimum when advancing to the next m_{real} row. At each row boundary this produces a jump $\Delta m_{\text{imag}} = m_{\text{imag,max}} - m_{\text{imag,min}}$, which can be much larger than the intra-row step.

The *snake-path* (boustrophedon) traversal eliminates this problem by reversing the sweep direction of m_{imag} on alternate rows:

$$\text{row } k: \quad m_{\text{imag}} = \begin{cases} m_{\text{imag,min}} \to m_{\text{imag,max}} & k \text{ odd}, \\ m_{\text{imag,max}} \to m_{\text{imag,min}} & k \text{ even}. \end{cases}$$

This ensures that consecutive grid points always differ by either Δm_{real} or Δm_{imag} (one intra-row step), never by the full range. For the typical CAS sweep grid (18×26, $m_{\text{real}} \in [1.55, 2.4]$, $m_{\text{imag}} \in [0.15, 1.4]$), the maximum step size drops from $|\Delta m| \approx 1.25$ (raster) to $|\Delta m| = 0.05$ (snake), a factor of 25 improvement at each of the 17 row boundaries.

GPU-accelerated batched solver with hybrid CPU parallelism. As of v0.3.0, an optional CUDA.jl extension provides GPU-accelerated parameter sweeps with hybrid CPU parallelism. For a fixed (aggregate, medium condition) group, the translation operator $\mathbf{A}$ depends only on geometry and is shared across all refractive index values. The GPU solver exploits this by processing B RI values simultaneously: it uploads the shared FFT translation data to GPU once, then solves B independent CBICG systems $(\mathbf{I} - \mathbf{T}_b\mathbf{A})\boldsymbol{x}_b = \mathbf{T}_b\boldsymbol{p}_{\text{inc}}$ in parallel, where only $\mathbf{T}_b$ (the Mie T-matrix) varies per RI.

For large aggregates ($N_p \gtrsim 500$), the CPU phases of each batch (preparation of RHS vectors and post-processing of solutions) dominate wall time and leave the GPU idle. This bottleneck is addressed by two complementary measures:

1. **Intra-group CPU parallelism**: batch preparation and post-processing are parallelised over the B batch items using `Threads.@threads`, reducing the CPU-phase time by a factor of C (number of Julia threads).

2. **Hybrid GPU+CPU dispatch**: groups with $N_p \leq$ `gpu_np_threshold` (default 500) are processed by CPU threads running *concurrently* with the GPU task via `Threads.@spawn`, so no CPU core sits idle while the GPU handles a large-aggregate batch.

Running with `julia -t auto` activates both mechanisms. Measured on an RTX A6000 with a 24-core CPU (CAS sweep, $N_p = 350$–2800, FP32 mode), the hybrid implementation achieves ~6 jobs/s versus ~0.8 jobs/s without threading — a factor of ~7× speedup.

A Float32 precision mode is available (`gpu_float32=true`): Mie coefficients and the RHS are computed in Float64 on CPU, while the iterative solver runs in ComplexF32 on GPU. The convergence tolerance is automatically relaxed to 10^{-5}. Validation against the Fortran reference for $N_S = 1000$ spheres across the full CAS RI range shows that FP32 errors are dominated by the FFT translation approximation, not by the reduced floating-point precision.

4.5 Incident plane wave coefficients

For z-axis incidence ($\beta = 0$, $\alpha = 0$), only the $m = \pm 1$ multipole components are non-zero due to the azimuthal symmetry of a circularly decomposed plane wave. In the lr_tran basis, the incident field coefficients for sphere i are:

$$p^{(i)}_{mn,p=1,q=1} = i^{\,n+1}\sqrt{\frac{2n+1}{2}}\,\delta_{m,+1}\,e^{iz_i}, \qquad p^{(i)}_{mn,p=2,q=1} = -i^{\,n+1}\sqrt{\frac{2n+1}{2}}\,\delta_{m,-1}\,e^{iz_i} \tag{19}$$

for the first incident polarisation ($q = 1$), and

$$p^{(i)}_{mn,p=1,q=2} = -i^{\,n+2}\sqrt{\frac{2n+1}{2}}\,\delta_{m,+1}\,e^{iz_i}, \qquad p^{(i)}_{mn,p=2,q=2} = -i^{\,n+2}\sqrt{\frac{2n+1}{2}}\,\delta_{m,-1}\,e^{iz_i} \tag{20}$$

for $q = 2$. The phase factor e^{iz_i} accounts for the spatial position of sphere i along the propagation direction.

5 Output Quantities

5.1 Translation to a common origin

After solving for the per-sphere scattered-field coefficients $\boldsymbol{a}^{(i)}$, all coefficients are re-expanded about a common origin $\boldsymbol{r}_0$ (typically the aggregate centroid) using regular (Bessel j_n) translation:

$$\boldsymbol{a}_0 = \sum_{i=1}^{N_S} \mathbf{J}_{0i}\, \boldsymbol{a}^{(i)} \tag{21}$$

where $\mathbf{J}_{0i}$ is the regular translation matrix from sphere i to the common origin. The translation order for each sphere is determined by a convergence test on the translation addition theorem (Fortran routine `tranordertest`), ensuring that the re-expansion captures the scattered field to a specified accuracy.

The common-origin lr_tran coefficients are then transformed from the $(p = 1, p = 2)$ basis to mode-indexed form:

$$a_{mn}^{(\text{mode}\,1)} = a_{mn}^{(p=1)} + a_{mn}^{(p=2)}, \qquad a_{mn}^{(\text{mode}\,2)} = a_{mn}^{(p=1)} - a_{mn}^{(p=2)}. \tag{22}$$

These mode-1 and mode-2 coefficients correspond to the standard TE and TM VSWF expansion of the total scattered field about the common origin (Mackowski 1994, §3).

5.2 Scattering amplitude matrix (BH83 convention)

The far-field scattered wave is described by the 2×2 amplitude matrix (BH83, Eq. 3.12):

$$\begin{pmatrix} E_{\parallel}^{\text{sca}} \\ E_{\perp}^{\text{sca}} \end{pmatrix} = \frac{e^{ikr}}{-ikr} \begin{pmatrix} S_2 & S_3 \\ S_4 & S_1 \end{pmatrix} \begin{pmatrix} E_{\parallel}^{\text{inc}} \\ E_{\perp}^{\text{inc}} \end{pmatrix} \tag{23}$$

where S_1, S_2, S_3, S_4 are dimensionless complex functions of the scattering angles (θ, ϕ).

For the forward ($\theta = 0°$) and backward ($\theta = 180°$) directions only the $m = \pm 1$ terms of the common-origin expansion survive. The amplitudes are computed from the mode coefficients as

$$S_j = -2 \sum_{n=1}^{N_{\max}} \sum_{m=\pm 1} \sum_{p=1}^{2} F_j(n, m, p)\, \tau_{m,n,p}(\theta) \tag{24}$$

where F_j selects the appropriate combination of mode coefficients and incident polarisation (mode-1 from $q = 1$, mode-2 from $q = 1$ for S_2, S_4; similarly with sign change from $q = 2$ for S_1, S_3), and $\tau_{m,n,p}(\theta)$ are angular functions derived from the Wigner d-matrix at $\cos\theta$.

At $\theta = 0°$ the angular functions simplify to (Mackowski & Mishchenko 2011, `taufunc` routine):

m	p	$\tau_{m,n,p}(0°)$
$+1$	1	$-f_n$
$+1$	2	$-f_n$
-1	1	$+f_n$
-1	2	$-f_n$

where $f_n = \frac{1}{4}\sqrt{(2n+1)/2}$.

At $\theta = 180°$ an additional factor of $(-1)^n$ appears, and the signs of the $p = 1$ vs $p = 2$ contributions differ:

m	p	$\tau_{m,n,p}(180^\circ)$
$+1$	1	$-(-1)^n f_n$
$+1$	2	$+(-1)^n f_n$
-1	1	$+(-1)^n f_n$
-1	2	$+(-1)^n f_n$

Note on S_3 and S_4: For a single sphere or a symmetric dimer with the symmetry axis along z, $S_3 = S_4 = 0$ by symmetry. For general aggregates lacking such symmetry, all four elements are non-zero.

5.3 Cross-section efficiencies

The dimensionless efficiency factors are defined as $Q = C/(\pi\, a_{\text{eff}}^2)$ where C is the corresponding cross section, and computed for unpolarised incidence by averaging over the two incident polarisations $q = 1, 2$.

Extinction efficiency (optical theorem, Mackowski 1994, Eq. 12):

$$Q_{\text{ext}} = -\frac{2}{x_{\text{eff}}^2} \sum_{q=1}^{2} \text{Re}\Big(\boldsymbol{a}_q^\dagger\, \boldsymbol{p}_{\text{inc},q}\Big) \tag{25}$$

where the inner product $\boldsymbol{a}_q^\dagger \boldsymbol{p}_{\text{inc},q}$ is evaluated in the per-sphere basis (before common-origin translation). This is the multi-sphere generalisation of the forward-scattering (optical) theorem: the extinction is determined entirely by the projection of the scattered-field coefficients onto the incident-field pattern. The factor of 2 (instead of 4) in the prefactor arises from averaging the cross section over the two incident orthogonal polarisation states ($q = 1, 2$).

Scattering efficiency (from the coherent common-origin expansion, Mackowski 1994, Eq. 13):

$$Q_{\text{sca}} = \frac{1}{x_{\text{eff}}^2} \sum_{q=1}^{2} \Big(\|\boldsymbol{a}_{0,q}^{(\text{mode}\,1)}\|^2 + \|\boldsymbol{a}_{0,q}^{(\text{mode}\,2)}\|^2 \Big) \tag{26}$$

where the norm is the sum of squared magnitudes of all common-origin expansion coefficients for incident polarisation q. This requires the common-origin re-expansion (Sec. 5, first subsection), which coherently accounts for the relative phases of all spheres.

Absorption efficiency:

$$Q_{\text{abs}} = Q_{\text{ext}} - Q_{\text{sca}} \tag{27}$$

following from energy conservation.

5.4 MI02 scattering amplitudes

The MI02 (Mishchenko, Travis & Lacis 2002) amplitude matrix $\mathbf{S}^{\text{MI02}}$ has dimension of length and is related to the dimensionless BH83 amplitudes by:

$$S_{11}^{\text{MI02}} = \frac{S_2^{\text{BH83}}}{-ik}, \quad S_{22}^{\text{MI02}} = \frac{S_1^{\text{BH83}}}{-ik}, \quad S_{12}^{\text{MI02}} = \frac{S_3^{\text{BH83}}}{ik}, \quad S_{21}^{\text{MI02}} = \frac{S_4^{\text{BH83}}}{ik} \tag{28}$$

where $k = 2\pi\, n_{\text{med}}/\lambda_0$ is the (dimensional) wavenumber in the medium. Note the index mapping: the MI02 S_{11} element corresponds to BH83 S_2 (not S_1), and vice versa.

5.5 CAS-v2 observable amplitudes

The Complex Amplitude Sensing (CAS-v2) protocol (Moteki & Adachi 2024) directly measures three complex amplitudes derived from the MI02 matrix elements:

$$S_s^{\mathrm{fwd}} = S_{11}^{\mathrm{MI02}} + i\, S_{12}^{\mathrm{MI02}} \tag{29}$$

$$S_p^{\mathrm{fwd}} = S_{22}^{\mathrm{MI02}} - i\, S_{21}^{\mathrm{MI02}} \tag{30}$$

$$S^{\mathrm{bak}} = \frac{-S_{11}^{\mathrm{MI02,bwd}} + S_{22}^{\mathrm{MI02,bwd}}}{\sqrt{2}} \tag{31}$$

where S_s^{fwd} and S_p^{fwd} are the forward scattering amplitudes for s- and p-polarised incident light, respectively, and S^{bak} is the depolarisation-sensitive backward scattering amplitude.

6 Computational Flow Summary

Step	Operation	Description	Source file
1	Non-dimensionalise	$x_i = k \cdot a_i$, $\boldsymbol{r}_i \to k \cdot \boldsymbol{r}_i$, $m_{\mathrm{rel}} = m_{\mathrm{sphere}}/n_{\mathrm{med}}$	`ParameterSweep.jl`
2	Mie coefficients	a_n, b_n for each sphere	`MieCoefficients.jl`
3	Build T-matrix	lr_tran basis, per sphere	`TMatrixSolver.jl`
4	Incident field	$\boldsymbol{p}_{\mathrm{inc}}$ ($m = \pm 1$ terms)	`TMatrixSolver.jl`
5	CBICG solve	$(\mathbf{I} - \mathbf{TA})\boldsymbol{x} = \mathbf{T}\boldsymbol{p}_{\mathrm{inc}}$; warm-start from previous m (§4.4)	`TMatrixSolver.jl`
6	Common origin	Translate coefficients	`ScatteringAmplitude.jl`
7	Amplitudes	S_1–S_4 at $\theta = 0°, 180°$	`ScatteringAmplitude.jl`
8	Efficiencies	Q_{ext}, Q_{sca}, Q_{abs}	`ScatteringAmplitude.jl`
9	MI02 & CAS	Convert amplitudes	`ParameterSweep.jl`

7 Implementation Mapping

Theory	Equation(s)	Implementation	File
Wavenumber, non-dim.	(1)–(2)	`compute_scattering`, `run_parameter_sweep`	`ParameterSweep.jl`
Mie coefficients a_n, b_n	(5)–(9)	`mie_coefficients`	`MieCoefficients.jl`
lr_tran T-matrix	(10)	`build_tmatrix`	`TMatrixSolver.jl`
Translation $\mathbf{H}_{ij}$	(14)–(15)	`translate_coefs`	`TranslationCoefs.jl`
FFT translation	(16)	`fft_translate`, `setup_fft_grid`	`FFTTranslation.jl`
CBICG solver	(13), (17), Table 1	`_solve_bicg`	`TMatrixSolver.jl`
GMRES solver (optional)	(13), (18), Table 2	`gmres` (Krylov.jl)	`TMatrixSolver.jl`
RI continuation (§4.4)	—	`initial_amn` kwarg	`TMatrixSolver.jl`, `ParameterSweep.jl`
Incident coefficients	(19)–(20)	`incident_coefficients`	`TMatrixSolver.jl`
Common-origin expansion	(21)–(22)	`translate_to_common_origin`	`ScatteringAmplitude.jl`
BH83 amplitudes S_1–S_4	(23)–(24)	`compute_amplitude_matrix`	`ScatteringAmplitude.jl`
Cross-section efficiencies	(25)–(27)	`compute_efficiencies`	`ScatteringAmplitude.jl`
MI02 conversion	(28)	`bh83_to_mi02`	`ParameterSweep.jl`
CAS-v2 observables	(29)–(31)	`read_results_h5_for_cas.jl`	`scripts/`
Aggregate I/O	—	`read_aggregate_file`, `read_aggregate_catalog`	`AggregateIO.jl`
GPU extension (optional, via `using CUDA`)			
GPU batched BiCG	(13), Table 1	`gpu_solve_bicg_batch!`	`ext/.../gpu_batch_solver.jl`
GPU FFT translation	(16)	`gpu_apply_A_batch!`	`ext/.../gpu_fft_translation.jl`
GPU T-matrix / S-parity	(10), (17)	`gpu_apply_T_batch!`	`ext/.../gpu_kernels.jl`

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